

Transducers

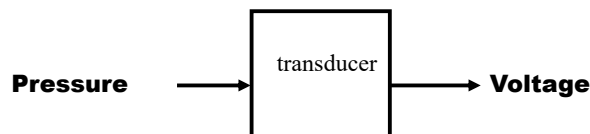
What has been covered in these slides : Transducer introduction, its block diagram and two main elements, its characteristics, various classification of transducers with some examples too, working of LVDT in the last.

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INTRODUCTION OF TRANSDUCERS

- A transducer is a device that converts one form of energy into another.
- In the context of engineering measurements, transducers are commonly used to convert physical quantities such as pressure, temperature, displacement, or flow into electrical signals or vice versa.
- In other word it is a device that is capable of converting the physical quantity into a proportional electrical quantity such as voltage or current. This voltage or current can be used for measuring the initial physical quantity.
- Transducers have vast applications in the field of measurements.
- Example :

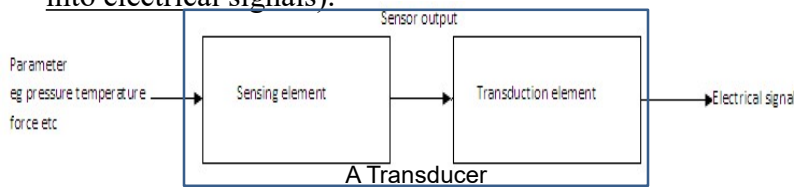


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BLOCK DIAGRAM OF TRANSDUCERS : Its main elements

- Transducer contains two main elements that are closely related to each other i.e. **the sensing element and transduction element**.
- The sensing element is called as the **sensor**. It is a device producing measurable response to change in physical conditions. (i.e. it reads the input signal).
- The transduction element convert the sensor output to suitable electrical form (i.e. it converts the sensor signal into electrical signals).



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CHARACTERISTICS OF TRANSDUCERS

1. **Ruggedness** : Mechanically strong, suitable for rough and tough use.
2. **Linearity** : Output vs input variation should be linear.
3. **Repeatability i.e., precision** : Same outputs should be obtained for same inputs, every time (precision).
4. **Accuracy** : Result or output should be very close to the expected or verified output.
5. **High stability and reliability** : Outputs should not deviate without any change in input, and the device should be trustworthy.
6. **Speed of response** : outputs should be attained as soon as the inputs are fed, without any time-lag.
7. **Sensitivity**: Outputs should be clear and visible even for smaller changes in inputs.
8. **Small size** : size of the device should be compact and small, so that it could be handled easily.

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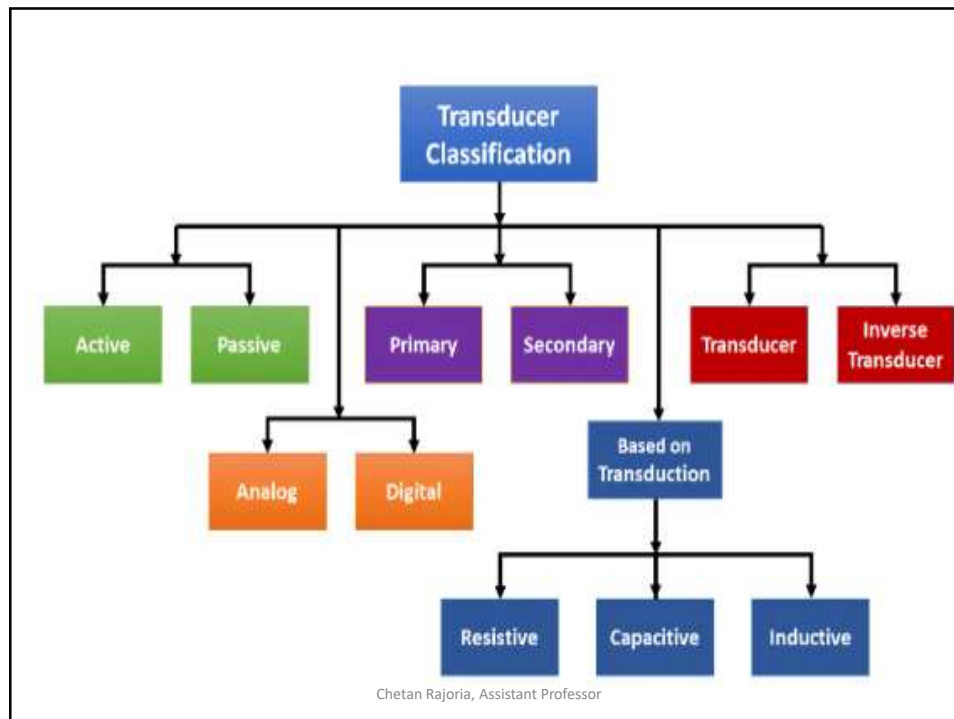
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CLASSIFICATION OF TRANSDUCERS

- The transducers can be classified according to their application, method of energy conversion, nature of the output signal, and so on.
- Following is the classification of transducers—
 - I. Active and passive transducers.
 - II. Analog and digital transducers.
 - III. On the basis of transduction principle used : Capacitive / Inductive / Resistive
 - IV. Primary and secondary transducer
 - V. Transducers and inverse transducers.

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ACTIVE AND PASSIVE TRANSDUCERS

- **Active transducers :**
- These transducers do not need any external source of power for their operation. No battery or cell is required to make it working. Therefore they are also called as self generating type transducers.
- I. The active transducer are self generating devices which operate under the energy conversion principle.
- II. As the output of active transducers we get an equivalent electrical output signal e.g. temperature or strain to electric potential, without any external source of energy being used.

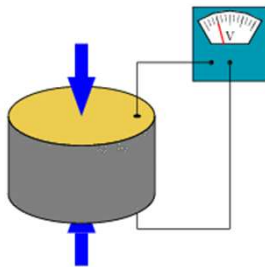
Example : A **Piezo-electric transducer** (details in next slide)

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Piezoelectric Transducer



This type of transducer can convert the amount of applied external load (or pressure) directly into the electrical signals, in proportional amount. As the load is applied, a proportional change in voltage can be obtained, which can be read on a volt meter. Here, no external battery is required to make it working.

Examples of artificial piezo-electric material : Barium Titanate, Lithium Niobate, PZT (Lead-Zirconate-Titanate).

Some natural piezoelectric materials : Quartz, Topaz, Tourmaline crystal

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- **Passive Transducers :**

- I. These transducers **need external source of power** for their operation. So they are not self generating type transducers.
- II. A DC power supply or an audio frequency generator is used as an external power source.
- III. These transducers produce the output signal in the form of variation in resistance, capacitance, inductance or some other electrical parameter in response to the quantity to be measured.

Example : **LVDT** (Linearly Variable Differential Transformer)
(used in Proving Ring for force measurement (*working details on slide no 19*))

Another example : **Strain gauge**.

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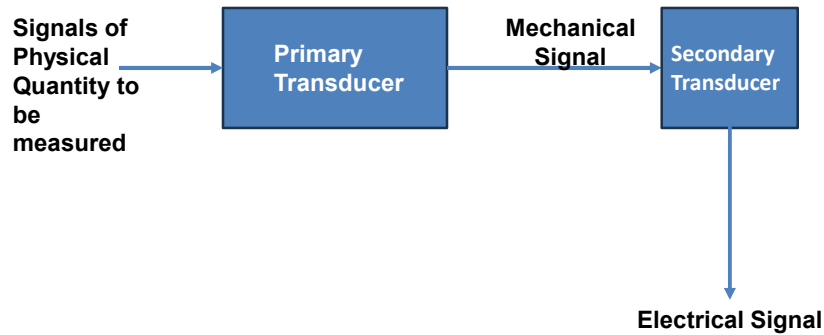
PRIMARY AND SECONDARY TRANSDUCERS

- Some transducers contain the mechanical as well as electrical device. The mechanical device converts the physical quantity to be measured into a mechanical signal. Such mechanical device are called as the primary transducers, because they deal with the physical quantity to be measured.
- The electrical device then convert this mechanical signal into a corresponding electrical signal. Such electrical device are known as secondary transducers. (*see the block diagram on the next slide*).
- Means, if *only one conversion of signal* is there, then the transducer is called primary type. Or if there are *two, three of more conversions of signals*; such transducers are called secondary type, tertiary type, etc.

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PRIMARY AND SECONDARY TRANSDUCERS



Example : In unbonded strain gauge arrangement, the diaphragm acts as primary transducer and wires of the strain gauge act as secondary transducer. (see the next slide).

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Analog and Digital Transducers

- **Analog transducers** produce a **continuous output signal** (usually voltage or current) that is directly proportional to the input physical quantity. The signal varies smoothly over a range. A continuous graph will be obtained if drawn on x-y coordinates. It is suitable for precise measurements. Examples : thermocouple, strain gauge, LVDT.
- **Digital transducers** produce an **output in the form of digital pulses or discrete binary values (0 and 1)**. They represent data in digital format rather than continuous signals (stepped or pulsing output will be obtained). Examples : digital temperature sensor, ultrasonic proximity sensor, digital tachometer.

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Transducers classification on the basis of transduction principle

- Transducers can be classified according to the principle of transduction—i.e., how they convert a physical quantity into an electrical signal. These are the main types-
- **Resistive Transducers** : Working Principle: Change in resistance due to change in physical quantity (length, temperature, strain, light, etc.). Example : strain gauge, RTD.
- **Capacitive Transducers** : Working Principle: Change in capacitance due to variation in distance between plates, area of plates, or dielectric constant. Examples : Capacitive level sensor, capacitive pressure sensor.
- **Inductive Transducers** : Working Principle: Change in inductance of a coil due to movement of a magnetic material or another coil, causing change of flux. Examples : LVDT, inductive proximity sensor.
- There are many other types of transduction (conversions) also, which overlap with various other classification categories too. Examples : **thermoelectric transducers, piezoelectric transducer, photo-voltaic transducer, etc.**

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Transducer and Inverse Transducer

TRANSDUCER:

- Transducers convert non electrical quantity to electrical quantity (e.g microphone : converts voice (sound) signals into electrical signals. There are many other examples too, like strain gauge, potentiometer, thermocouples, solar cells, etc.)

INVERSE TRANSDUCER:

- Inverse transducers, also known as 'Actuators' convert electrical quantity to some other physical quantity (e.g loudspeaker : converts electrical signals into sound signals), electrical motor, pneumatic cylinders

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THANK YOU

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